

Reader: Obviously, the sequence $\left(\frac{\Delta f(x_0)}{\Delta x_n}\right)$ converges to $f'(x_0)$. In other words, the limit of $\left(\frac{\Delta f(x_0)}{\Delta x_n}\right)$ is the derivative of f at x_0 .

Author: What are the grounds for this conclusion?

Reader: Why, isn't it self-evident?

Author: Let me help you. Your conclusion is based on definition 2 of the limit of function at a point. Don't you think so?

Reader: Yes, I agree. Indeed, a certain number (in this case $f'(x_0)$) is the limit of a function $\phi(\Delta x)$ (in this case $\phi = \frac{F}{G}$) at $\Delta x = 0$ if for any sequence (Δx_n) convergent to zero the corresponding sequence $(\phi(\Delta x_n))$ converges to this number. Sequence (3) is precisely the sequence $(\phi(\Delta x_n))$ in our case.

Author: Correct. We have thus found that $\lim_{n \rightarrow \infty} \tan \alpha_n = f'(x_0)$. Now look at Fig. 38 and tell me which direction is the limit for the sequence composed of the chords $AB_1, AB_2, AB_3, \dots, AB_n, \dots$?

Reader: It is the direction of the tangent to the graph $f(x)$ at point $x = x_0$.

Author: Correct. Denote the slope of the tangent line by α_0 . Thus $\lim_{n \rightarrow \infty} \tan \alpha_n = \tan \alpha_0$ Consequently, $f'(x_0) = \tan \alpha_0$

We thus obtain the following geometrical interpretation of the derivative:
The derivative of a function f at a point x_0 is defined by the slope of the tangent to the graph of the function f at the point $x = x_0$.

